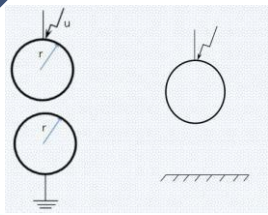
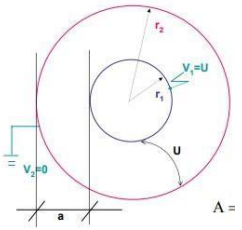
EE4047 – High Voltage Techniques I

New Topic:
Spherical Electrode Systems





Week 9

Asst. Prof. Dr. M. Alparslan

21.04.2026

OUTLINE

1 Concentric Spherical Electrode Systems

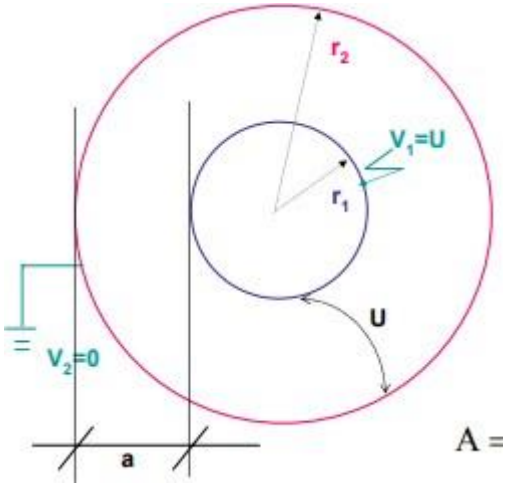
2 Multilayer Dielectric Concentric Spheres

1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

An electrode system that consists of two metalic spheres that has the same center, with a dielectric material between them.

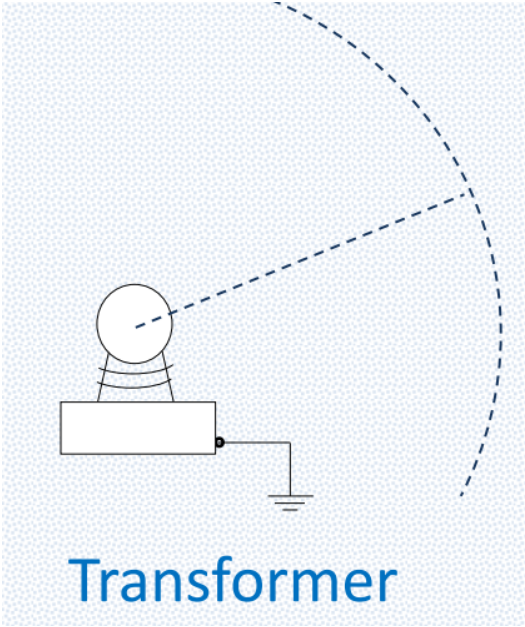
Although it does not have direct applications in the field, the shape and positioning of many electrodes require design and analysis assuming a concentric spherical electrode systems.

Spherical electrodes or electrodes which have shape similar to a sphere are widely used for reducing electric field strength in an area or take precautions for electric discharge events.



1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

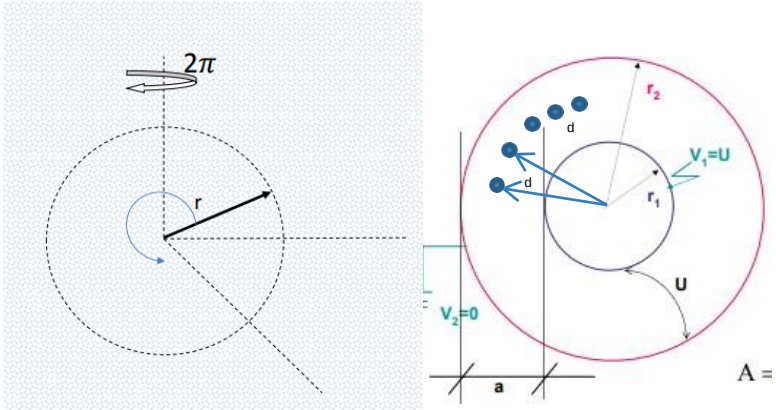
Considering such systems as a concentric spherical electrode system which has an outer electrode with a very large radius simplifies the solutions.



1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

Due to rotational symmetry of sphere, potential V which changes in 3 dimensions can be analyzed in one dimension with respect to radius r, the distance from the center of the system.

$$\frac{d^2V}{dr^2}+\frac{2}{r}\frac{dV}{dr}=0 \qquad V=A+\frac{B}{r}$$



Solving A and B constants using the boundary conditions as $V = V_1 = U$ for $r = r_1$, and $V = V_2 = 0$ foí $r = r_2$

$$V = U\frac{r_1}{r_2 - r_1}\left(\frac{r_2}{r} - 1\right)$$

Negative gradient of which gives us electric field strength:

$$E = U\frac{r_1.r_2}{r_2 - r_1}.\frac{1}{r^2}$$

1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

Due to symmetricity of the spherical electrode system, the flux density leaving the surface of a spherical electrode with r radius can be calculated by replacing the S in the equation

$$D = \frac{Q}{S}$$

with the equation that gives the surface of a sphere.

$$D = \frac{Q}{4\pi r^2}$$

Electric field strength can be derived using the relation $D = \varepsilon E$

$$E = \frac{Q}{4\pi r^2}\frac{1}{r^2}$$

Integral of elecric field strength equation gives us the voltage applied across the two concentric spherical electrodes:

$$U = \frac{Q}{4\pi\varepsilon}\frac{r_2 - r_1}{r_1.r_2} \quad \leftrightarrow \quad \frac{Q}{4\pi\varepsilon} = U\frac{r_1.r_2}{r_2 - r_1}$$

1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

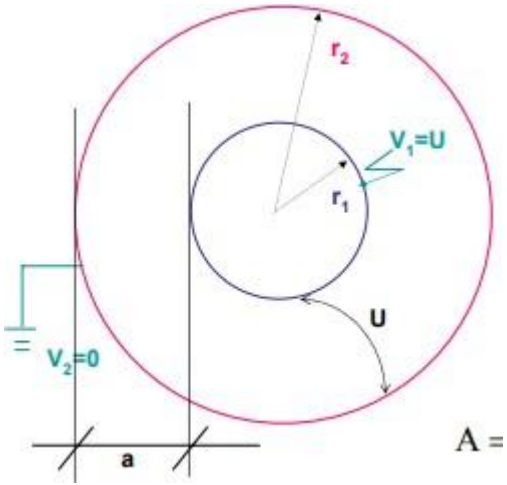
Using the equation for electric field strength, the maximum, minimum and average electric field strengths can be found.

The general electric field strength formula we derived in the previous slides:

$$E = U\frac{r_1.r_2}{r_2 - r_1}.\frac{1}{r^2} \quad \rightarrow \quad E = E_{ma} = \frac{U}{\frac{r_1}{r_2}(r_2 - r_1)}$$

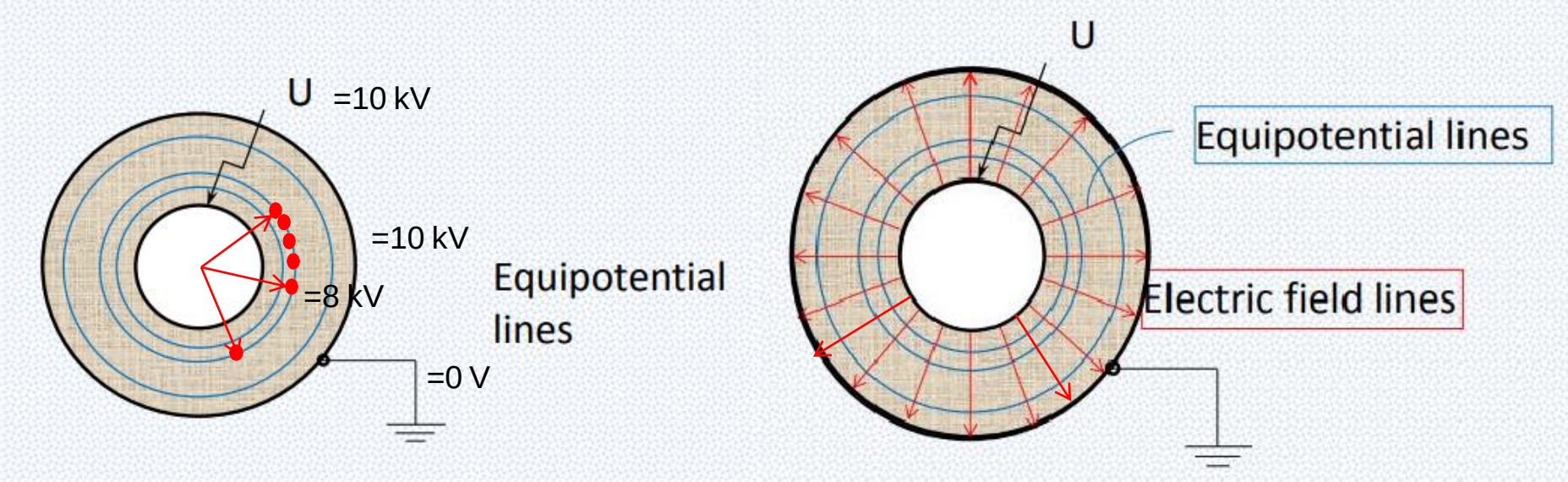
$$r = r_2 \quad \rightarrow \quad E = E_{mi} = \frac{U}{\frac{r_2}{r_1}(r_2 - r_1)}$$

$$r = r_{av} = \sqrt{r_1.r_2} \quad \rightarrow \quad E = E_{av} = U\frac{r_1.r_2}{r_2 - r_1}.\frac{1}{r_{ave}{}^2} = \frac{U}{r_2 - r_1}$$

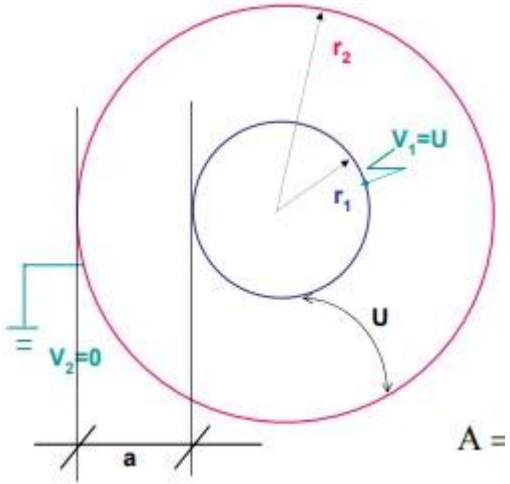


1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

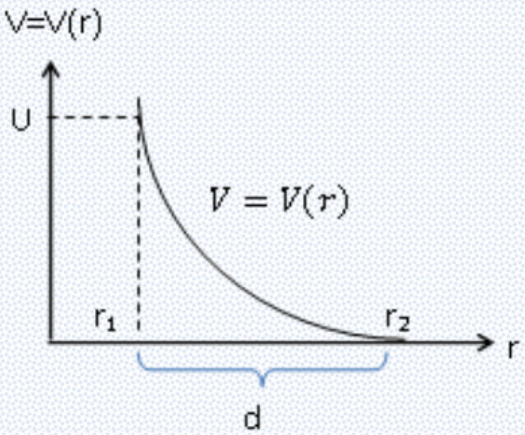
Electric field lines ALWAYS perpendicular to equipotential lines (ORTHOGONAL).

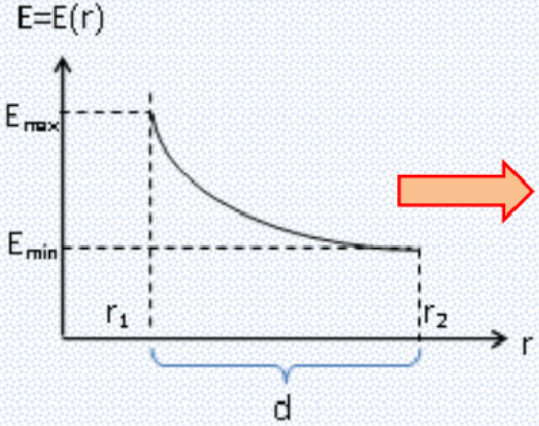


1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS



A =





Does it look uniform to you?

1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

Capacity of a concentric spherical electrode system can be calculated using the voltage equation we previously derived using the sphere surface area and charge. The unit for the inner shpere radius r_1 and out sphere radius r_2 should be (m), dielectric constant unit should be (F/m). $\varepsilon_0 = 8.854.10^{-12}\ F/m$ constant can be used and the result will be in Farad.

$$U = \frac{Q}{4\pi\varepsilon}\frac{r_2 - r_1}{r_1.r_2}$$

$$C = \frac{Q}{U} = 4\pi\varepsilon\frac{r_1.r_2}{r_2 - r_1} = 4\pi\varepsilon_r\varepsilon_0\frac{r_1.r_2}{r_2 - r_1}$$

1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

When defining, calculating and comparing different spherical electrode systems, we use a number of parameters.

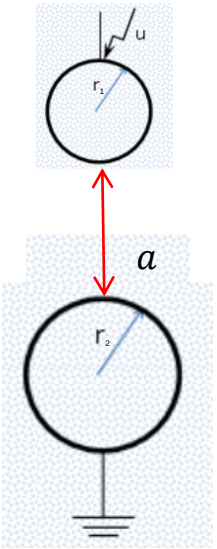
Actual electrode seperation, $a = r_2 - r_1$

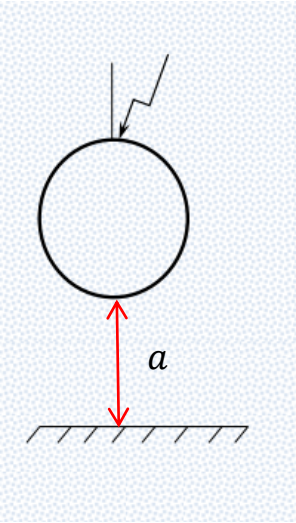
Effective electrode seperation, $\alpha = \frac{U}{E_{max}} = \frac{r_1}{r_2}(r_2 - r_1)$

Geometric characteristics, $p = \frac{r_1 + a}{r_1} = \frac{r_2}{r_1}$ $\quad q = \frac{r_2}{r_1} = p$

For concentric SES

Utilization factor (electrode efficiency), $\eta = \frac{\alpha}{a} = \frac{\frac{r_1}{r_2}(r_2 - r_1)}{(r_2 - r_1)} = \frac{r_1}{r_2} = \frac{1}{p}$





CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

If utilization factor (η), actual electrode seperation (a) and the voltage applied to the system are known, E_{max} can be calculated through

$U = E_{max}.\alpha = E_{max}.a.\eta$

If $E_{max} > E_{breakdown}$ the discharge starts closer to the spherical electrode with the smaller radius.

In nonuniform electric fields, breakdown does not happen immediately. Primarily partial discharges take place. As voltage increases these discharges become more frequent and after passing a number of stages full breakdown is encountered. This process depends on the level of nonuniformity of the field.

Level of nonuniformity (inverse of utilization factor): $\frac{1}{\eta}$ (in concentric SES, equal to p)

OUTLINE

1 Continuation to: Concentric Spherical Electrode Systems

2 Multilayer Dielectric Concentric Spheres

1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

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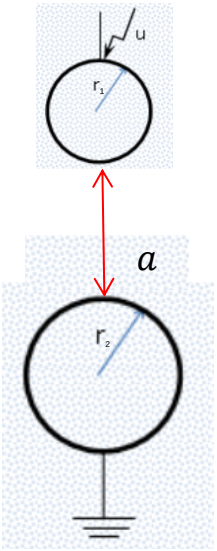
Actual electrode seperation, $a = r_2 - r_1$

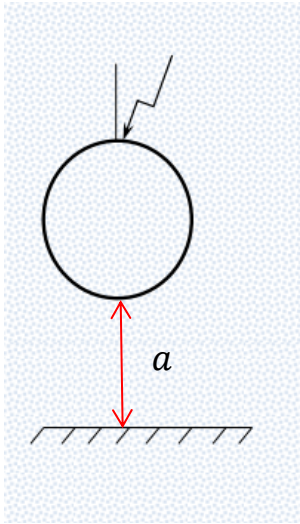
Effective electrode seperation, $\alpha = \frac{U}{E_{max}} = \frac{r_1}{r_2}(r_2 - r_1)$

Geometric characteristics, $p = \frac{r_1 + a}{r_1} = \frac{r_2}{r_1}$ $\qquad q = \frac{r_2}{r_1} = p$

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Utilization factor (electrode efficiency), $\eta = \frac{\alpha}{a} = \frac{\frac{r_1}{r_2}(r_2 - r_1)}{(r_2 - r_1)} = \frac{r_1}{r_2} = \frac{1}{p}$





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Level of nonuniformity (inverse of utilization factor): $\frac{1}{\eta}$ (in concentric SES, equal to p)

1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

Two characteristic cases in breakdown behavior of spherical electrode systems:

Case 1: Radius of the outer sphere (r_2) is constant, applied voltage (U) is constant, change of E_{max} with inner sphere radius (r_1)

$$\frac{d(E_{max})}{dr_1} = \frac{d\left[U\frac{r_2}{r_1}\frac{1}{r_2 - r_1}\right]}{dr_1} = -\frac{(r_2 - 2r_1).r_2.U}{(r_1.r_2 - {r_1}^2)} = 0$$

$$r_1 = \frac{r_2}{2}$$

This is the case that maxes E_{max} minimum

$$(E_{max})_{min} = U\frac{r_2}{\frac{r_2}{2}}\left[\frac{1}{r_2 - \left(\frac{r_2}{2}\right)}\right] = \frac{4U}{r_2} = \frac{2U}{r_1} = 2.E_{ave}$$

The optimum geometric characteristic, utilization factor and effective electrode seperation:

$p_b = \frac{r_1 + a}{r_1} = \frac{r_2}{r_1} = 2$ $\eta_b = \frac{1}{p_b} = 0{,}5$ $\alpha_{max} = \frac{a}{2}$

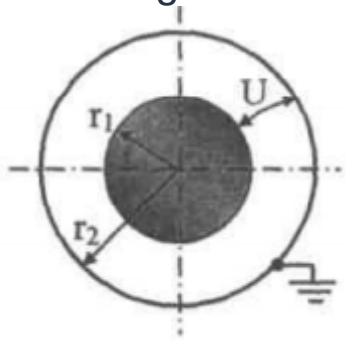
1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

Example-1: For a concentric spherical electrode system, the inner sphere radius is given as 8 mm.

a) Determine the ideal outer sphere radius for the optimum geometric characteristic.

Solution-a: $\frac{r_2}{r_1} = 2 \qquad r_2 = 2.r_1 = 2.8 = 16\,mm = 1{,}6\,cm$

b) If an insulation material with breakdown strength $E_d = 100\,kV/cm$ and dielectric constant $\varepsilon_r = 4$ is used in the given system, calculate the maximum applicable voltage and system capacitance. ($\varepsilon_0 = 8{,}854.10^{-12}\,F/m$)

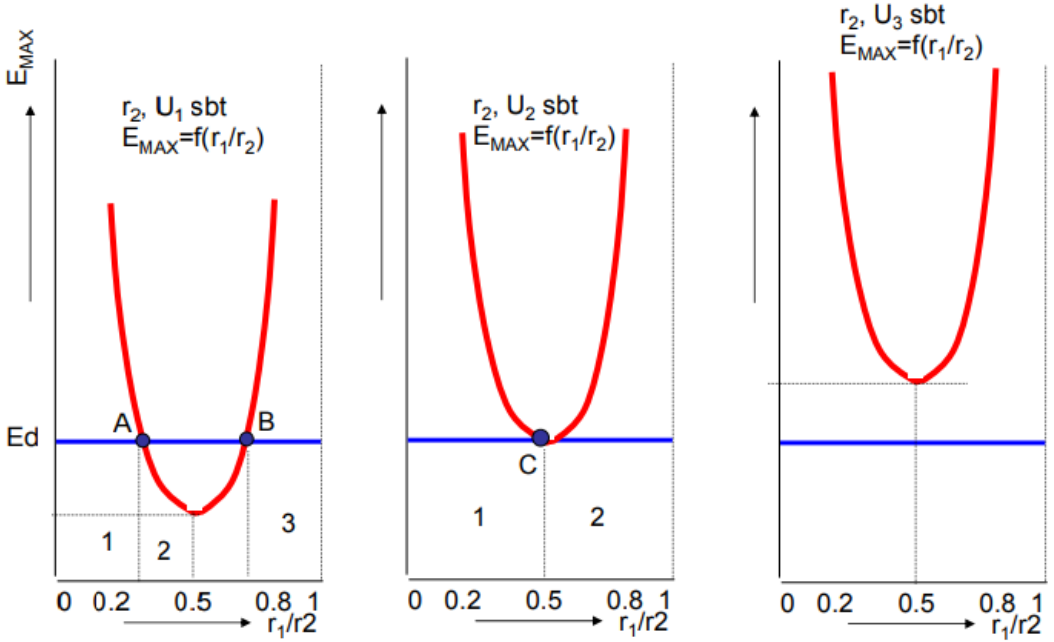


Solution-b: $U_d = E_d \frac{r_1}{r_2}(r_2 - r_1) = 100.\frac{0{,}8}{1{,}6}(1{,}6 - 0{,}8) = 40\,kV$

$$C = \frac{4.\pi.\varepsilon.r_1.r_2}{r_2 - r_1} = \frac{4.\pi.8{,}854.10^{-14}.4.1{,}6.0{,}8}{1{,}6 - 0{,}8} = 7{,}12.10^{-12}\,F = 7{,}12\,pF$$

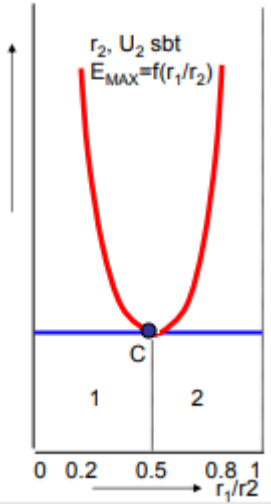
1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

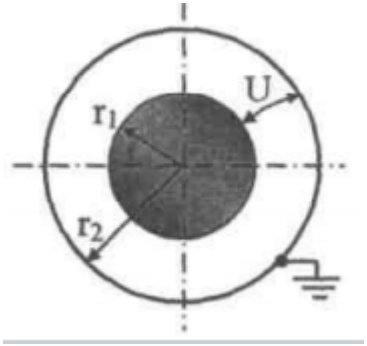
Breakdown analysis for different constant breakdown strength (E_b) and different applied voltage (U) values



CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

Example-2: For a concentric spherical electrode system, the outer sphere radius is given as 200 cm. For inner sphere radius 50 cm and 120 cm and air breakdown strength of 30 kV/cm, calculate the voltage value that starts discharges and explain how discharges happen.





Solution-2: The voltage that starts discharges

$$U_b = E_b . \frac{r_1}{r_2}(r_2 - r_1)$$

For $r_1 = 50\ cm$, $U_b = 30 . \frac{50}{200} . (200 - 50) = 1125\ kV$

For $r_1 = 120\ cm$, $U_b = 30 . \frac{120}{200} . (200 - 120) = 1440\ kV$

Geometric characteristics for each case:

For $r_1 = 50\ cm$, $p = \frac{200}{50} = 4$ $\frac{r_1}{r_2} = 0{,}25$

For $r_1 = 120\ cm$, $p = \frac{200}{120} = 1{,}66$ $\frac{r_1}{r_2} = 0{,}6$

2 MULTILAYER DIELECTRIC CONCENTRIC SPHERES

This dielectric configuration is connected in series.

U is the voltage applied to the entire system

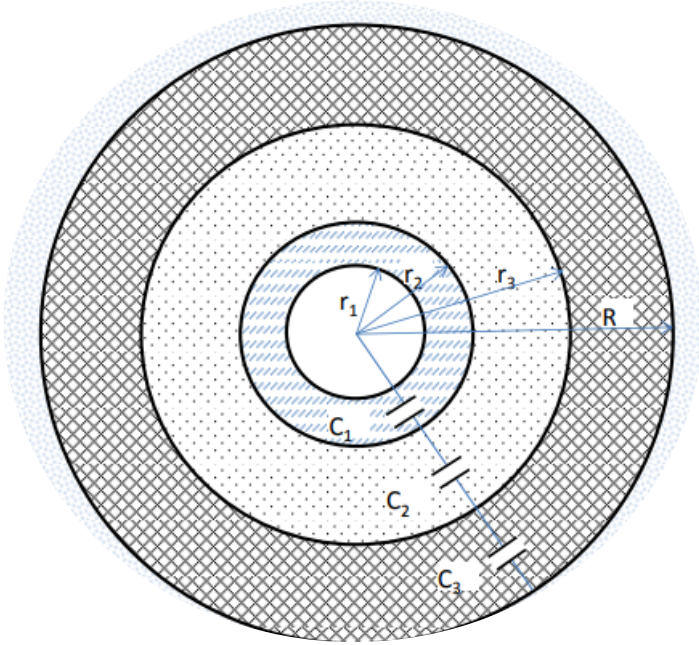
U_1, U_2, …, U_n are voltages across the corresponding dielectric layers

$$I = I_1 = I_2 = I_3 = \cdots = I_n$$

$$Q = Q_1 = Q_2 = Q_3 = \cdots = Q_n$$

$$C\,U = C_1U_1 = C_2U_2 = C_3U_3 = \cdots = C_nU_n$$

$$U = U_1 + U_2 + U_3 + \cdots + U_n$$



2 MULTILAYER DIELECTRIC CONCENTRIC SPHERES

This dielectric configuration is connected in series.

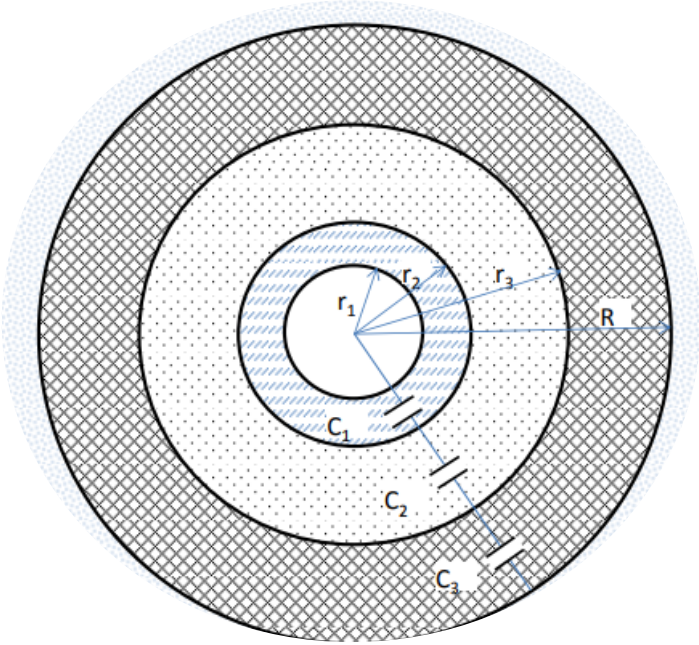
$$C\,U = C_1U_1 = C_2U_2 = C_3U_3 = \cdots = C_nU_n$$

$$U_1 = \frac{C\,U}{C_1}$$

$$U_2 = \frac{C\,U}{C_2}$$

$$\vdots$$

$$U_n = \frac{C\,U}{C_n}$$



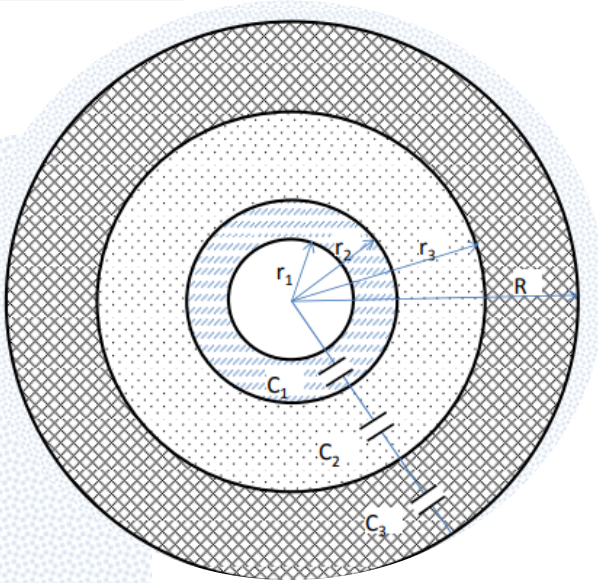
2 MULTILAYER DIELECTRIC CONCENTRIC SPHERES

This dielectric configuration is connected in series.

$$X_C = X_{C1} + X_{C2} + X_{C3} + \cdots + X_{Cn}$$

$$\frac{1}{\omega C} = \frac{1}{\omega C_1} + \frac{1}{\omega C_2} + \frac{1}{\omega C_3} + \cdots + \frac{1}{\omega C_n}$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_n}$$



C : equivalent capacitance of the system, total capacitance of the system

U : voltage applied to the entire system

C_i : capacitance of the ith dielectric layer

U_i : voltage across i^{th} dielectric layer

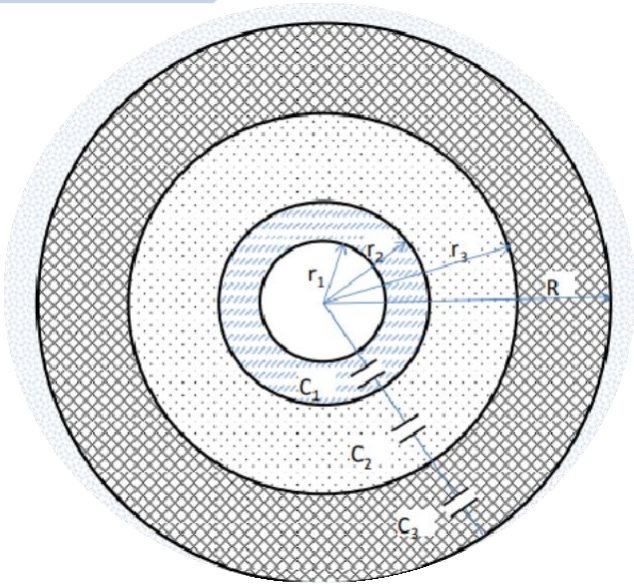
2 MULTILAYER DIELECTRIC CONCENTRIC SPHERES

$$C_1 = 4\pi\varepsilon_1 \frac{r_1 r_2}{r_2 - r_1}, C_2 = 4\pi\varepsilon_2 \frac{r_2 r_3}{r_3 - r_2}, \dots, C_n = 4\pi\varepsilon_n \frac{r_n R}{R - r_n}$$

$$\frac{1}{C} = \frac{1}{4\pi\varepsilon_1 \frac{r_1 r_2}{r_2 - r_1}} + \frac{1}{4\pi\varepsilon_2 \frac{r_2 r_3}{r_3 - r_2}} + \cdots + \frac{1}{4\pi\varepsilon_n \frac{r_n R}{R - r_n}}$$

$$= \frac{1}{4\pi\varepsilon_0}\left(\frac{1}{\varepsilon_{r1}}\frac{r_2 - r_1}{r_1 r_2} + \frac{1}{\varepsilon_{r2}}\frac{r_3 - r_2}{r_2 r_3} + \cdots + \frac{1}{\varepsilon_{rn}}\frac{R - r_n}{r_n R}\right)$$

$$= \frac{1}{4\pi\varepsilon_0}\underbrace{\sum_{i=1}^{n} \frac{1}{\varepsilon_{ri}}\frac{r_{i+1} - r_i}{r_i\, r_{i+1}}}_{A}$$

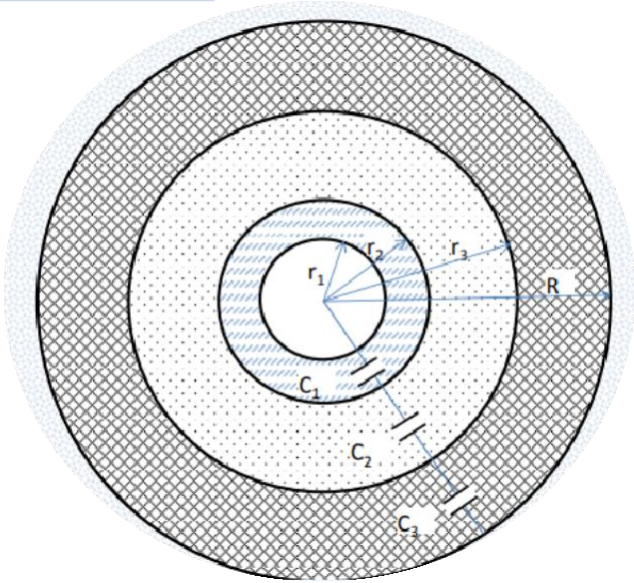


2 MULTILAYER DIELECTRIC CONCENTRIC SPHERES

This dielectric configuration is connected in series.

$$A = \sum_{i=1}^{n} \frac{1}{\varepsilon_{ri}} \frac{r_{i+1} - r_i}{r_i\, r_{i+1}} = \frac{1}{\varepsilon_{r1}} \frac{r_2 - r_1}{r_1 r_2} + \frac{1}{\varepsilon r_2} \frac{r_3 - r_2}{r_2 r_3} + \cdots + \frac{1}{\varepsilon_{rn}} \frac{R - r_n}{r_n R}$$

$$\frac{1}{C} = \frac{A}{4\pi\varepsilon_0} \Rightarrow C = \frac{4\pi\varepsilon_0}{A}$$

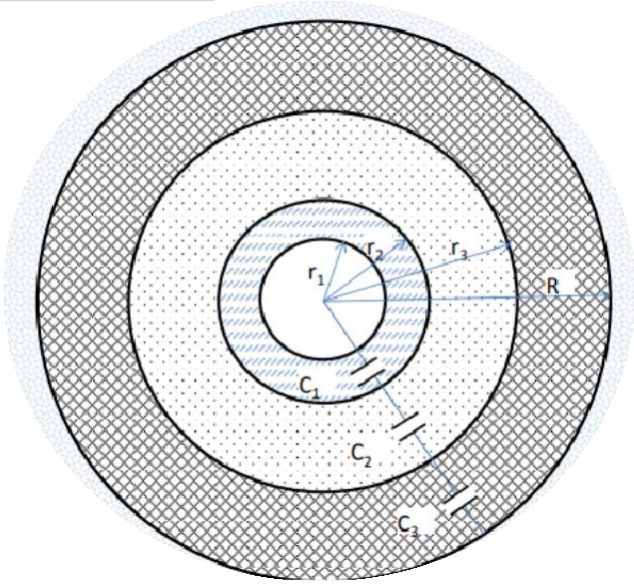


2 MULTILAYER DIELECTRIC CONCENTRIC SPHERES

$$U_1 = \frac{C\,U}{C_1} = \frac{\frac{4\pi\varepsilon_0}{A}U}{4\pi\varepsilon_0\varepsilon_{r1}\frac{r_1 r_2}{r_2 - r_1}} = \frac{U}{A\varepsilon_{r1}}\frac{r_2 - r_1}{r_1 r_2}$$

$$E_{1max} = \frac{U_1}{\frac{r_1}{r_2}(r_2 - r_1)}$$

$$E_{1min} = \frac{U_1}{\frac{r_2}{r_1}(r_2 - r_1)}$$



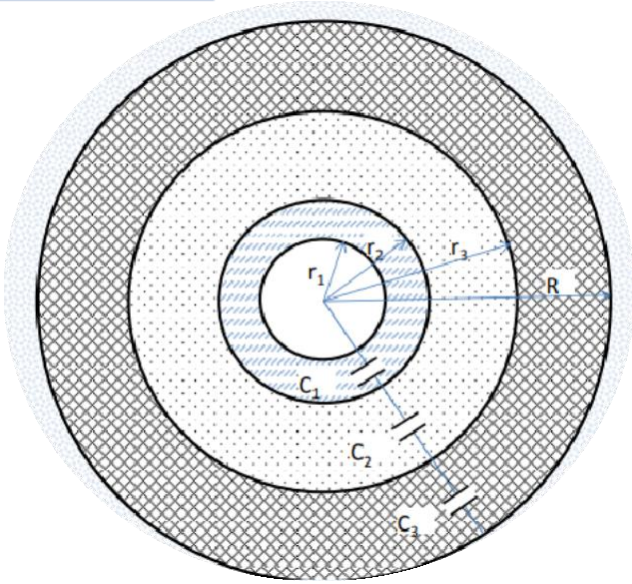
2 MULTILAYER DIELECTRIC CONCENTRIC SPHERES

$$U_2 = \frac{C\,U}{C_2} = \frac{\frac{4\pi\varepsilon_0}{A}U}{4\pi\varepsilon_0\varepsilon_{r2}\frac{r_2 r_3}{r_3 - r_2}} = \frac{U}{A\varepsilon_{r2}}\frac{r_3 - r_2}{r_2 r_3}$$

$$E_{2max} = \frac{U_2}{\frac{r_2}{r_3}(r_3 - r_2)} = \frac{\frac{U}{A\varepsilon_{r3}}\frac{r_3 - r_2}{r_2 r_3}}{\frac{r_2}{r_3}(r_3 - r_2)} = \frac{U}{Ar_2^2\varepsilon_{r3}}$$

$$E_{2min} = \frac{U_2}{\frac{r_3}{r_2}(r_3 - r_2)} = \frac{\frac{U}{A\varepsilon_{r3}}\frac{r_3 - r_2}{r_2 r_3}}{\frac{r_3}{r_2}(r_3 - r_2)} = \frac{U}{Ar_3^2\varepsilon_{r3}}$$

We can write the same equations with respect to applied voltage (U)

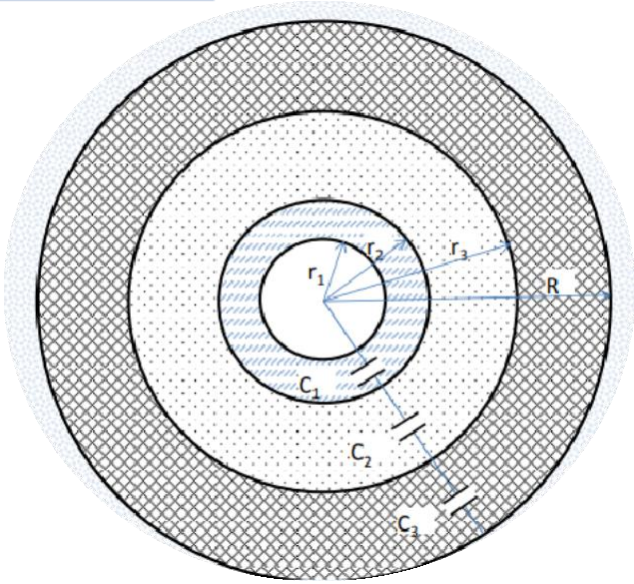


2 MULTILAYER DIELECTRIC CONCENTRIC SPHERES

$$U_i = \frac{C\,U}{C_i} = \frac{\frac{4\pi\varepsilon_0}{A}U}{4\pi\varepsilon_0\varepsilon_{ri}\frac{r_i r_{i+1}}{r_{i+1}-r_i}} = \frac{U}{A\varepsilon_{ri}}\frac{r_{i+1}-r_i}{r_i\,r_{i+1}}$$

$$E_{imax} = \frac{U_i}{\frac{r_i}{r_{i+1}}(r_{i+1}-r_i)}$$

$$E_{1min} = \frac{U_i}{\frac{r_{i+1}}{r_i}(r_{i+1}-r_i)}$$

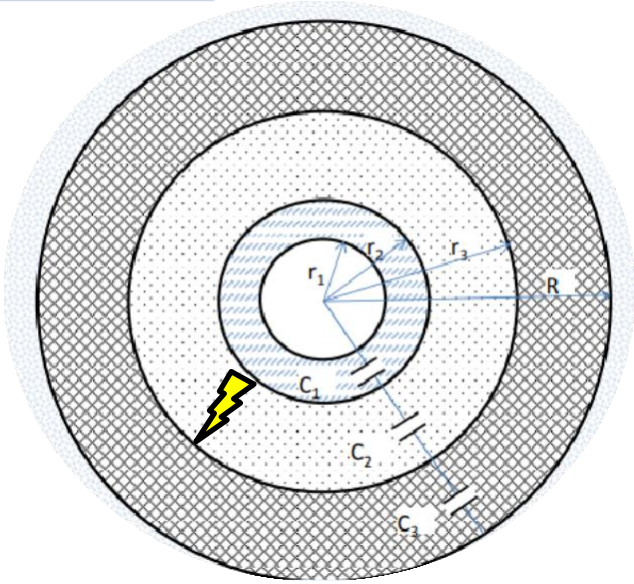


2 MULTILAYER DIELECTRIC CONCENTRIC SPHERES

Electric field strength across any dielectric layer, E_i, becomes greater than breakdown strength of that dielectric, there will be an electrical discharge.

$E_{i,max} \geq E_{bi} \rightarrow$ Discharge in that layer of the insulation!!!

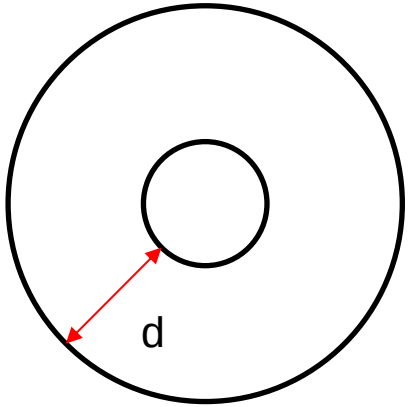
Breakdown, short circuit in i^{th} dielectric layer

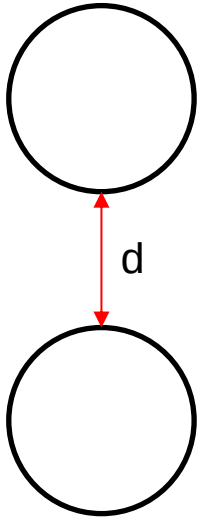


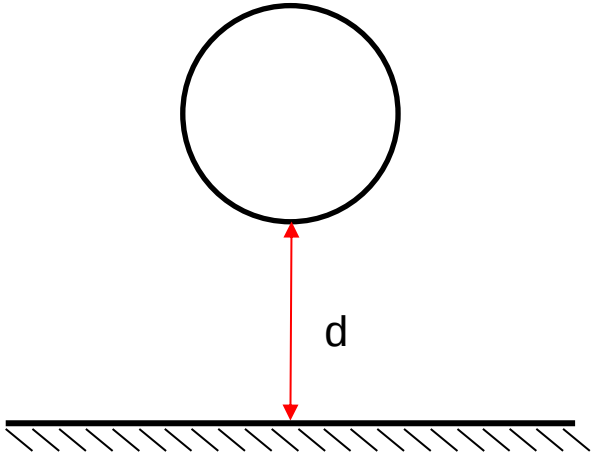
3 BREAKDOWN BEHAVIOR OF SPHERICAL ELECTRODE SYSTEMS

Actual electrode separation or insulator thickness, d

$$E_{ave} = \frac{U}{d} \rightarrow d = \frac{U}{E_{ave}}$$

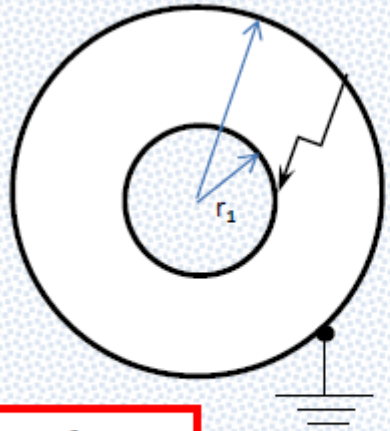




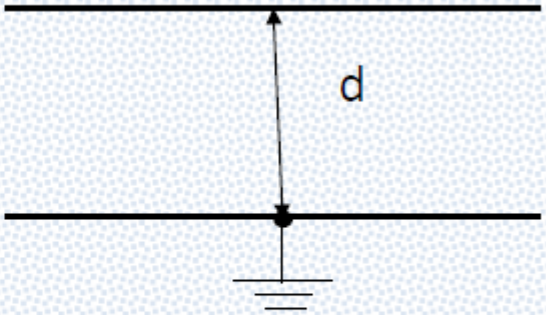


3 BREAKDOWN BEHAVIOR OF SPHERICAL ELECTRODE SYSTEMS

$$\eta = \frac{E_{ave}}{E_{max}}$$

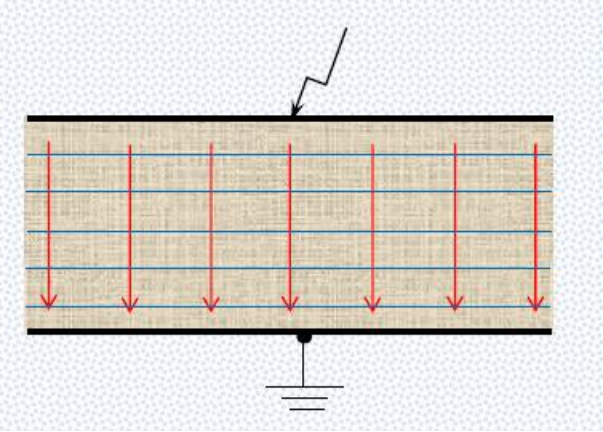


Non-uniform electric field

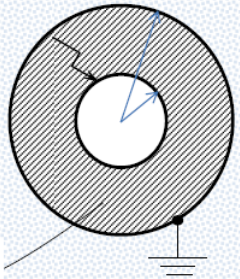


uniform electric field

3 BREAKDOWN BEHAVIOR OF SPHERICAL ELECTRODE SYSTEMS

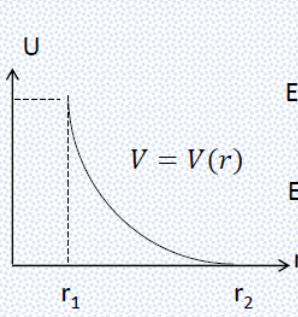


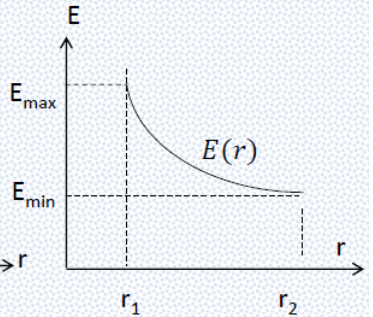
$$E = \frac{U}{d} = E_{ave} = E_{max} = E_{min}$$



$V = V(r)$

$E = E(r)$





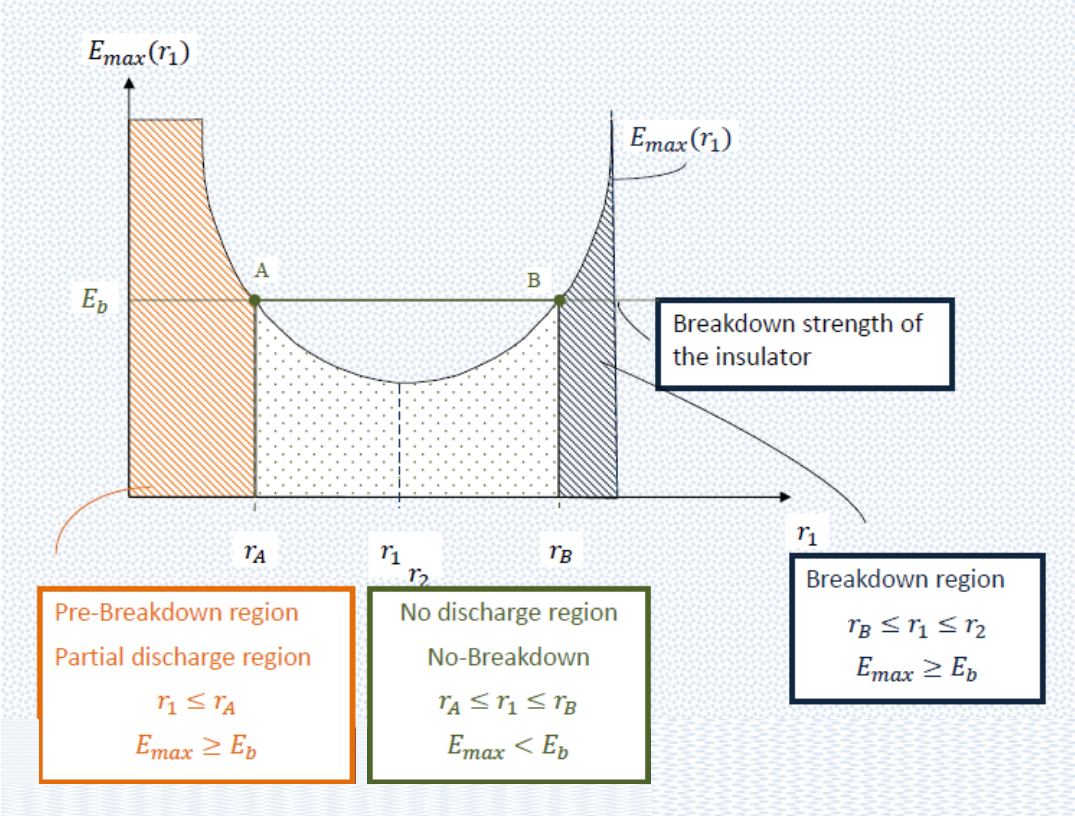
- Change in E_{max} with respect to r_1
- Change in U_{max} with respect to r_1

$$\frac{r_2}{r_1} = p = 2$$

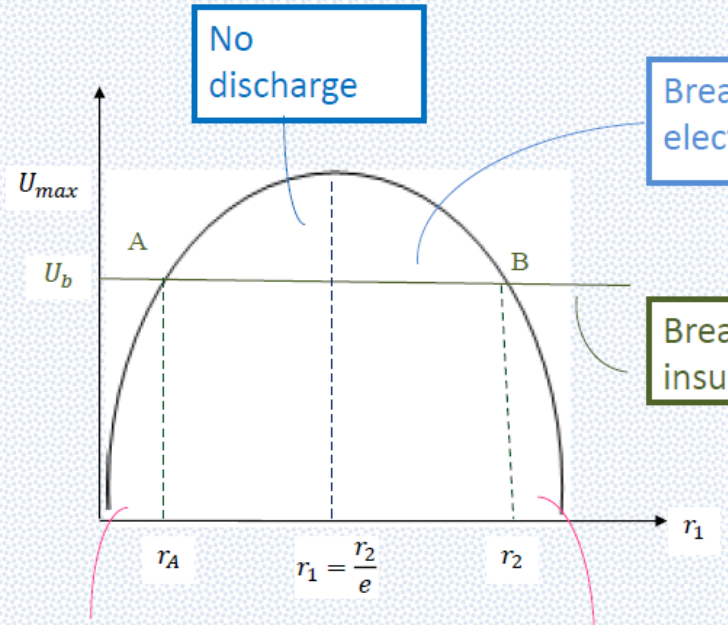
$p = p_b$

fittest, optimum geometric char. with regard to breakdown in concentric spheres

3 BREAKDOWN BEHAVIOR OF SPHERICAL ELECTRODE SYSTEMS



3 BREAKDOWN BEHAVIOR OF SPHERICAL ELECTRODE SYSTEMS



$$U_{bsys} > U_{binsulator} \rightarrow No\ dicharge$$

$$U_{bsys} \leq U_{binsulator} \rightarrow Dicharge$$

3 BREAKDOWN BEHAVIOR OF SPHERICAL ELECTRODE SYSTEMS

- $r_2 \rightarrow constant, E_b \rightarrow constant$
- $E_{max} = \frac{U}{\frac{r_1}{r_2}(r_2 - r_1)} = \frac{U}{\alpha}$
- $U = \alpha . E_{max}$

- $\alpha = \alpha_{max}, U = U_{max}, E_{max} = E_b$

$$U_{max} = \alpha_{max} . E_b$$

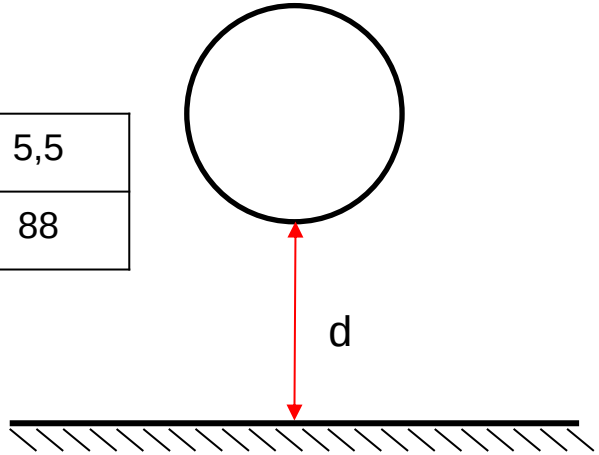
$$\alpha = \frac{r_1}{r_2}(r_2 - r_1) = f(r_1)$$

Maximum → Voltage will be **Maximum**

3 BREAKDOWN BEHAVIOR OF SPHERICAL ELECTRODE SYSTEMS

Example-1: In the sphere-plane electrode system shown in the figure, the spherical electrode Radius r=10 mm.
The breakdown voltages for the tests made with different electrode seperations are given in the table. (take k as 0,95).

d (mm)	5	4,5	4,8	5,1	4,7	5,5
$U_d\ (kV)$	81	75	80	86	76	88



a) Calculate the electric breakdown strength for the given values
b) Calculate the average electric breakdown strength
c) Find the minimum electric breakdown strength

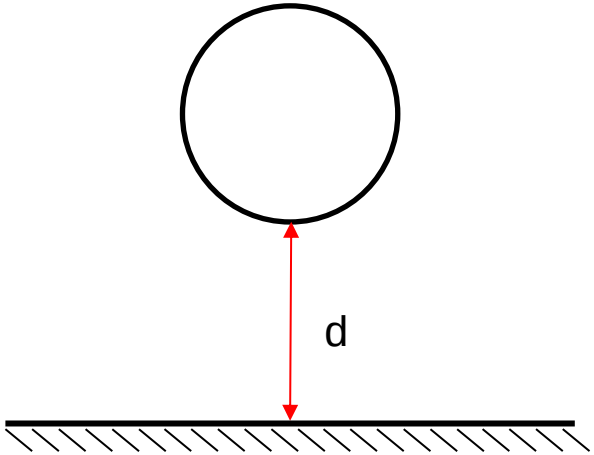
Solution:

a) For concentric spherical electrode system our equation was

$$E_{max} = \frac{U}{\left(\frac{r_1}{r_2}\right).(r_2 - r_1)}$$

For sphere-plane electrode system, replacing r_1 with r and r_2 with $(r + d)$

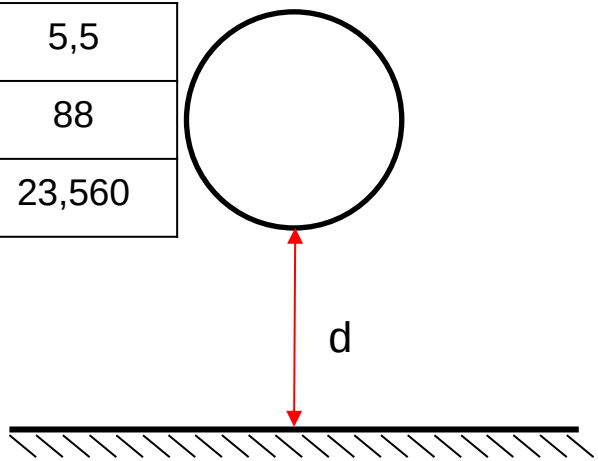
$$E_{max} = k\frac{U}{\left(\frac{r}{r+d}\right).(r + d - r)} = k\frac{U}{\frac{r.d}{r+d}} = k\frac{r+d}{r.d}U$$



d (mm)	5	4,5	4,8	5,1	4,7	5,5
$U_d\ (kV)$	81	75	80	86	76	88
$E_d\ (kV/mm)$	23,085	22,958	23,433	24,190	22,582	23,560

Solution:

d (mm)	5	4,5	4,8	5,1	4,7	5,5
$U_d\ (kV)$	81	75	80	86	76	88
$E_d\ (kV/mm)$	23,085	22,958	23,433	24,190	22,582	23,560



b)

$$E_{breakdown-ave} = \frac{23{,}085 + 22{,}958 + 23{,}433 + 24{,}190 + 22{,}582 + 23{,}560}{6} = 23{,}301\,\frac{kV}{mm}$$

$$= 233{,}01\ kV/cm$$

c) $E_{breakdown-min} = 22{,}582\,\frac{kV}{mm} = 225{,}82\ kV/cm$

END OF THE LESSON

Any questions?

Check out our playlist s on Youtube:

High Voltage Technique: https://www.youtube.com/playlist?list=PLLZI7WaEmZi04KMWVURGlU34W3LoNZyWX

Insulators: https://www.youtube.com/playlist?list=PLLZI7WaEmZi1ReLGq0qfP1VjRqxDJi5kA

Lightning Impulse Voltage: https://www.youtube.com/playlist?list=PLLZI7WaEmZi2isWOB-zBbb6LNplH28byQ